

Commerce XI Maths App

Activities 2.1–2.4

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 21

Question 1 - Activity 2.1 · Step 1

For daily sugar sales 120, 75, 33, 140.5, 50 and 70.5 kg, identify L.

Correct answer: 140.5

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 140.5.

Question 2 - Activity 2.1 · Step 2

For daily sugar sales 120, 75, 33, 140.5, 50 and 70.5 kg, identify S.

Correct answer: 33

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 33.

Question 3 - Activity 2.1 · Step 3

For daily sugar sales 120, 75, 33, 140.5, 50 and 70.5 kg, Calculate the range in kg.

Correct answer: 107.5

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 107.5.

Question 4 - Activity 2.2 · Step 1

For 2, 4, 5, 6, 8, 10, 12, 14, 20, 30, 60, Enter the position number of Q_1 .

Correct answer: 3

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 3.

Question 5 - Activity 2.2 · Step 2

For the ordered data 2, 4, 5, 6, 8, 10, 12, 14, 20, 30, 60, Enter Q_1 .

Correct answer: 5

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 5.

Question 6 - Activity 2.2 · Step 3

For the ordered data 2, 4, 5, 6, 8, 10, 12, 14, 20, 30, 60,
Enter the position number of Q_3 .

Correct answer: 9

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 9.

Question 7 - Activity 2.2 · Step 4

For the ordered data 2, 4, 5, 6, 8, 10, 12, 14, 20, 30, 60,
Enter Q_3 .

Correct answer: 20

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 20.

Question 8 - Activity 2.2 · Step 5

For the ordered data 2, 4, 5, 6, 8, 10, 12, 14, 20, 30, 60,
Enter the interquartile range.

Correct answer: 15

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 15.

Question 9 - Activity 2.3 · Step 1

Returns for four years are ₹1000, ₹3000, ₹4500 and ₹5000.
Find the mean.

Correct answer: 3375

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 3375.

Question 10 - Activity 2.3 · Step 2

Returns for four years are ₹1000, ₹3000, ₹4500 and ₹5000.

The deviations $x - \bar{x}$ are

Options

A. -2375, -375, 1125, 1625

B. 2375, 375, -1125, -1625

C. -3375, -375, 1125, 5000

D. -2375, 375, 1125, 1625

Correct answer: -2375, -375, 1125, 1625

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is -2375, -375, 1125, 1625.

Question 11 - Activity 2.3 · Step 3

Returns for four years are ₹1000, ₹3000, ₹4500 and ₹5000.

The squared deviations are

Options

A. 5640625, 140625, 1265625, 2640625

B. 2375, 375, 1125, 1625

C. 5640625, 140625, 1125625, 2640625

D. 5640625, 140625, 1265625, 2460625

Correct answer: 5640625, 140625, 1265625, 2640625

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 5640625, 140625, 1265625, 2640625.

Question 12 - Activity 2.3 · Step 4

For returns ₹1000, ₹3000, ₹4500 and ₹5000,

Find $\Sigma(x - \bar{x})^2$.

Correct answer: 9687500

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 9687500.

Question 13 - Activity 2.3 · Step 5

For returns ₹1000, ₹3000, ₹4500 and ₹5000,

Find $\text{Var}(X)$.

Correct answer: 2421875

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 2421875.

Question 14 - Activity 2.3 · Step 6

For returns ₹1000, ₹3000, ₹4500 and ₹5000,
Find S.D. correct to two decimal places.

Correct answer: 1556.24

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 1556.24.

Question 15 - Activity 2.4 · Step 1

The first student group records prices of

Options

- A. rice
- B. wheat
- C. sugar
- D. tur-dal

Correct answer: rice

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is rice.

Question 16 - Activity 2.4 · Step 2

The second student group records prices of

Options

- A. wheat
- B. rice
- C. sugar
- D. tur-dal

Correct answer: wheat

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is wheat.

Question 17 - Activity 2.4 · Step 3

The third student group records prices of

Options

- A. sugar
- B. rice
- C. wheat
- D. tur-dal

Correct answer: sugar

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is sugar.

Question 18 - Activity 2.4 · Step 4

The fourth student group records prices of

Options

- A. tur-dal
- B. rice
- C. wheat
- D. sugar

Correct answer: tur-dal

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is tur-dal.

Question 19 - Activity 2.4 · Step 5

Each group records its item price in

Options

- A. 10 shops
- B. 4 shops
- C. 5 shops
- D. 20 shops

Correct answer: 10 shops

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is 10 shops.

Question 20 - Activity 2.4 · Step 6

For each grocery item, every group finds

Options

- A. mean and standard deviation
- B. median only
- C. range only
- D. mode only

Correct answer: mean and standard deviation

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is mean and standard deviation.

Question 21 - Activity 2.4 · Step 7

The prices collected for rice, wheat, sugar and tur-dal may also be used to calculate

Options

- A. partition values
- B. only the largest value
- C. only the smallest value
- D. no statistical value

Correct answer: partition values

Explanation and steps

Follow the activity step in sequence, using range, quartiles, variance or standard deviation as requested. The required result is partition values.